

PAUL NAMALOMBA

🎓 **MSc(Eng) Structural, BSc(Hons.) Civil**

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RESUME SUMMARY

- Over **4 years of total technical experience** in Computational science and Digital infrastructure initiatives in Civil Engineering research and teaching at UCT, including end-to-end deployment of Computational Modelling software, usage of HPC clusters (Slurm, Rocky Linux). Further industry experience in building APIs in C#, Python and Javascript. Which includes:
- **2.5 years and counting** of project maintenance, and full-stack development of **SESKA finite-element computational suite** written in GNU C++, Fortran and GiD. First of its kind port from Linux to Windows whilst maintaining powerful Multiprocessing (Hyperthreading) compute methods.
- **4 months experience** developing industry-grade APIs in **C#** and **Javascript**, developing **Flutter** apps for Android, including a new ride-sharing app called Ecoride.
- **1.2 years' experience** architecting, maintaining, and optimizing the **SEAT EBE enrollment platform** using Python Django with HTML, PostgreSQL, and LDAP for Security Management – We protected against SQL injections by using the Python Django layer for additional security.
- Over **750 hours of active training** in Data Science, Data Engineering, High Performance Computing and Back-End Development.
- Led cross-functional teams of up to five teaching assistants in **lecturing** undergraduate students—aligning curriculum with emerging technologies and strategic learning objectives.
- Built my own python data processing packages utilizing Computer Vision for Visual Data Analysis with Google's Gemma AI models, Random Forest outlier detection and other machine-learning innovations. (
- My entire portfolio is available online at m website: (<https://paulnamalomba.github.io>).

POINTS OF REFERENCE

- My entire portfolio is available online at my website: (<https://paulnamalomba.github.io>)
- At my website I maintain guides on technical implementation and system file server management, as well as a history of all the technologies I have worked on **Web-Apps, Android Apps, Static Websites, Dashboards** and **APIs**.
- My proof of qualification is attached to this application.

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CORE TECHNICAL SKILLS

- **Programming Languages:** C++, C#, React Typescript, Python, SQL, Flutter, HTML, Javascript (Nuxt, Node)
- **Web Scraping, Data Science and Statistical Data Analysis:** Python, Jupyter(*), Microsoft Power BI, Microsoft Tableau, Microsoft Excel, Python RESTful APIs
- **RDBMS, ETL Pipelines and Big Data (Cloud Computing):** Azure Database with PostgreSQL, MySQL, SQLite, MongoDB, REDIS and Cloudflare R2 Object Storage.
- **Machine-Level Computer Programming:** Microsoft VSC++, GNU Autotools, Cygwin64, Mingw32/64 Cross-Compat Tools
- **Programming IoT/Realtime Devices and REST APIs:** Arduino-based MCUs, API programming with C# .NET and Python Django, SignalR Websockets
- **DevOps and CI/CD:** GitHub, Bitbucket, Vercel, DockerHub and Cloudflare.
- **High-Performance Computing and Parallel Computing:** HPC, Slurm Workload Manager, OpenMPI, PETSc, METIS.
- **Data Security Policies:** LDAP Configuration, BCrypt, JWT Bearer
- **Computational Modelling and Analysis:** SESKA, GiD, ABAQUS
- **Machine Learning and Regression Analysis:** Scikit-Learn, TensorFlow, statsmodels, datashadric
- **Advanced Shell Scripting, File System and Server Management and Automation:** Windows CMD Prompt, PowerShell 7+, Git BASH, MSYS2 Shells, Linux BASH.

OTHER TECHNICAL+RESEARCH SKILLS

- **Scientific Research Design, Reporting and Methodology**
- **The Engineering Design Process Workflow**
- A Multi-Disciplinary Depth of Knowledge (**Computing, Data Science and Statistics, Engineering and Research Management**)
- **Graphics Design for Posters:** Adobe Photoshop, Canva.
- **Other Advanced ICT Skills:** Microsoft Office, LibreOffice, Microsoft OneNote.
- **Operating Systems:** Linux Ubuntu 20.04+, Windows 10+, WSL; Linux Ubuntu Server 20.04+,

RESEARCH INTERESTS

- **Computational Mechanics and Applied Mathematics**
- **Shelf-Ice Mechanics and Rheology**
- **Non-linear Large-Deformation Elasticity, Visco-Plasticity, and Elasto-Plasticity**
- **Quasi-Brittle and Ductile Failure Modes**
- **Computational Algorithm Optimization**
- **Data Analysis and Digital Infrastructures**
- **Google Maps APIs and REST APIs**
- **Emerging Financial Technologies**
- **High-Performance and Parallel Computing**

PROFESSIONAL AND ACADEMIC EXPERIENCE

Professional: Chief Innovation Office and Lead Backend Engineer

ComputeMore Africa Technologies Ltd (Tech Startup) | Blantyre, Malawi

October 2025 – Present | Experience: 4 months

***As Chief Innovation Officer I am tasked with consulting on Systems Design, proposing new projects and the double role of Lead Backend Engineer gives me first-class authority over Systems Architecture, Quality Control, Implementation and DevOps**

- **Achievements:**
 - Developed the Ecoride App API (a new ride-sharing app in Malawi looking to implement interesting Financial Technologies with Mobile Money Airtel and TNM). **Stack:** Flutter, Vite Node.js JavaScript, Redis PostgreSQL, C# .NET ASPCORE, Google Maps API, PayChangu API, RabbitMQ and SignalR WebSocket
 - Developed new app called INTEGRATE for a Hotel ERMS (Enterprise Resource Management System) – **Stack:** Flutter, React Typescript, PostgreSQL, Redis, C# .NET ASPCORE, SignalR Websockets
 - Leveraging **Cloudflare R2 Object Storage** for egress-free, low-cost cloud storage
 - Built RestAPIs including Authentication, and Authorisation using **BCrypt** and **JWT** 256-bit tokens for security hardening. Clean architecture following the Controller -> DTOs -> Services -> Entities philosophy.
 - Fully managed DevOps with MS Azure and Render for Databases, Vercel for the web admin portals and our company website.
 - Our tech partners included Namecheap, Cloudflare for web hosting and cloud storage (R2); PayChangu (Fintech provider), and ICTAM.

Academic: SESKA* Computational Engineer as part of Doctor of Philosophy: Civil Engineering

University of Cape Town | Cape Town, South Africa

September 2023 – Present | Experience: 2.5 Years

***SESKA is an in-house Finite Element Analysis software package maintained in the Department of Civil Engineering**

Back-End: C++, and Fortran. **Client Computing:** PETSc. **HPC and Parallel Computing:** OpenMPI, UCT HPC.

- **Tasks and Achievements:**
 - Managed the project through Git and CI/CD Protocols.
 - Deployed SESKA onto UCT HPC, the EBE Computational Mechanics Labs, and Chemical Engineering Computer Labs.
 - Implemented of working **elasto-plasticity, visco-plasticity, material damage, cyclic loading, and variable elasticity** models.

- Full-Cycle development of windows Port of SESKA with UCRT and MVSC++, **the first of its kind.**

Academic: Assistant Lecturer in Engineering Mechanics – Experience: 2 Years

University of Cape Town | Cape Town, South Africa

February 2025 – July 2025 | June 2024 – December 2024 | June 2023 – December 2023 | June 2022 – December 2022

• Tasks and Achievements:

- 85%+ Course Pass Rate for CIV2011F (2025), 93%+ Course Pass Rate for CIV1007S (2022, 2023, 2024)
- Crafted the course outlines, course content and exams from Scratch.
- Led and directed a team of five Teaching Assistants, managed budgets and schedules
- Enhanced technical and practical instruction of students using **SESKA**

SEAT* Backend Application Developer (Hybrid) – Experience: 1.2 Years

University of Cape Town | Cape Town, South Africa

December 2025 – February 2026 | November 2024 – July 2025 | December 2021 – January 2022

***SEAT is an in-house enrolment platform developed at the Faculty of Engineering & the Built Environment**

Front-end: Python Django with HTML Views. **Database Architecture:** PostgreSQL (psycopg2) - **Remote Server-Hosted.**

• Tasks and Achievements:

- Code inspection, debugging, re-factoring and module linking (Python Django, JavaScript, HTML)
- Relational database editing, version control and management (MySQL, PostgreSQL, PostgreSQL Admin Tools)
- Improving overall functionality of key tools: Curriculum Selection, Curriculum Form automation and Student Timetable Management
- User verification and authorization protocols with LDAP for PeopleSoft
- Web administration and relational database server administration
- Email automation with SMTP and Python

EDUCATION HISTORY

Ongoing Doctor of Philosophy: Civil Engineering

University of Cape Town | Cape Town, South Africa | September 2023 – **Present** | Expected Completion **August 2026**

- **Topic of study** – *A Nonaffine Damage Theory for Multisolid Composites*
- **Research Keywords:** Composite Materials, Viscoelasticity, Computational Mechanics, Applied Mechanics
- **Skills Developed:** Research Management, Research Design and Reporting, HPC, Microsoft Visual Studio C++, GNU C++, GNU Fortran, Mingw32/64 Cross-Compatibility Tools, Python for Data Science, Git, MSYS2 Shells and Linux BASH.

Master of Science in Engineering: Structural Engineering & Materials

University of Cape Town | Cape Town, South Africa | March 2020 – December 2021

- **Distinction in Thesis** – *Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.*
- **Research Keywords:** Material Damage, Computational Mechanics, Applied Mechanics
- **Skills Developed:** Research Management, Research Design and Reporting, HPC, GNU C+, Python for Data Analysis, Git and Linux BASH.

Bachelor of Science (Honours): Civil Engineering

University of Cape Town | Cape Town, South Africa | January 2016 – December 2019

- **Distinction in Dissertation** – *Design of A Climate Chamber for Uni-axial Compressive Strength Testing of Laboratory Grown Sea Ice.*
- **Dean's Merit List:** 2016, 2017.

LICENSES & CERTIFICATES

Ongoing Professional Certificate: IBM Data Engineering

Coursera | July 2025 – **Present** | Expected Completion **End February 2026** | 8/16 Modules Completed (**150+ hours**)

- IBM – Introduction to Data Engineering
- IBM – Python for Data Science, AI & Development
- IBM – Python Project for Data Engineering
- IBM – Introduction to Relational Databases (RDBMS)
- IBM – Databases and SQL for Data Science with Python
- IBM – Hands-on Introduction to Linux Commands and Shell Scripting
- IBM – Relational Database Administration (DBA)
- IBM – Introduction to Big Data with Spark and Hadoop

Ongoing Professional Certificate: Microsoft Back-End Developer

Coursera | September 2025 – Present | Expected Completion End April 2026 | 3/8 Modules Completed (40+ hours)

- Microsoft – Foundations of Coding Back-End
- Microsoft – Security and Authentication
- Microsoft – Database Integration and Management

Professional Certificate: Google Advanced Data Analytics

Coursera | December 2024 – May 2025 | 300+ hours of training

- Google - Foundations of Data Science
- Google - Regression Analysis: Simplify Complex Data Relationships
- Google - The Power of Statistics
- Google - Go Beyond the Numbers: Translate Data into Insights
- Google - Get Started with Python
- Google - The Nuts and Bolts of Machine Learning

Developer Certificate: Data Analysis with Python

Coursera | April 2025 – May 2025 | 300+ hours of training

- Data Structures: Pandas dataframes, Numpy Arrays
- Data Cleaning and Visualization: Matplotlib, Seaborn
- Data Scraping and Data Extraction: .csv, HTML, BigQuery

ORGANIZATIONS AND TECH SUBSCRIPTIONS

- **GitHub Pro Contributor** (Open-Source and Closed-Surface) – 500+ contributions in last 6 months – <https://github.com/paulnamalomba>
- **PyPI Contributor** (kabwenzenamalomba@gmail.com – diversecelllar)
- **AfricasTalking API Subscription** (kabwenzenamalomba@gmail.com), **Airtel Africa API Developer Platform** (kabwenzenamalomba@gmail.com)
- **Upstash Context7 API Subscription** (kabwenzenamalomba@gmail.com)
- **Android Studio Developer Platform** (kabwenzenamalomba@gmail.com); **Google Cloud Developer Platform** (kabwenzenamalomba@gmail.com); **Google AI Pro Subscription** (kabwenzenamalomba@gmail.com)
- **South African Statistical Association**, July 2025 – Present [ID 2748543]
- **Statistics Without Borders**, January 2025 – Present [ID 16089]
- **South African Institution of Civil Engineering**, 2021 – Present [ID 202100183]

CONFERENCES & PUBLICATIONS

- **MSc Thesis 2022** [available: <http://hdl.handle.net/11427/37632>]: *Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.*
- **Africomp7 conference** – scheduled to present two abstracts:
 - **A Lie Group Approach to Non-Affine Interaction and Damage in Multi-Solid Composites** with **Prof. Sebastian Skatulla**, **Prof. Michael Kaliske**, **Prof. Carlo Sansour**
 - **Logarithmic Strain-Based Glen's Flow Law for Glaciological Applications** with **Prof. Sebastian Skatulla**, **Prof. Michael Kaliske**, **Prof. Carlo Sansour**, **Dr. Maxime Nutte**
- **ICMM7 Conference**, November 2022.
- **SACAM2020 Conference**, November 2021.

COMMUNITY INVOLVEMENT & LEADERSHIP

- **CRC Cape Town (Malawi): Service in Altar Team**, July 2025 – Present.
- **Redeemed For a Purpose Ministries (Malawi): Service in Financial Coordination**, April 2025 – Present.
- **International Students Forum (UCT):** March 2025 – Present.
- **Statistics Without Borders Volunteer**, January 2025 – Present.
- **Mentorship and Supervision of Honour's Researchers (UCT)**, June 2021 – Present.
- **Volunteer Work for IAPO Soccer Tournament (UCT)**, October 2022.
- **Flood and Disaster Relief Efforts (Mangochi, Malawi)**, January 2022.
- **Child Education Advocacy (Mangochi, Malawi)**, January 2017 & January 2018.

REFEREES

Prof. Sebastian Skatulla^[1,2] – PhD Research Supervisor, MSc Research Supervisor, SEAT Lead Developer

[1] Professor of Structural Engineering & Materials | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

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Dr. Maxime Nutte – Research Collaborator

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Mr. Thanthwe Kalonga^{1,2]} – Managing Director of ComputeMore Africa Technologies Ltd

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